QUBO Formulation: Maximum Clique Problem

Maximum Clique Problem

Problem Statement. Let $G = (V, E)$ be an undirected graph with vertex set V and edge set E. A clique is a subset $C \subseteq V$ such that every pair of distinct vertices in C is connected by an edge in E. The Maximum Clique Problem seeks a clique C^* of maximum cardinality, i.e., $|C^*| = \max\{|C| : C \subseteq V \text{ and } \forall u, v \in C, \{u, v\} \in E\}$.

Decision Variables. For each vertex $i \in V$, we introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i equals 1 if vertex i is included in the candidate clique, and 0 otherwise.

QUBO Derivation. Step 1 — The original objective is to maximize the number of selected vertices:

$$\max \sum_{i \in V} x_i. \tag{1}$$

Since QUBO solvers minimize energy, we negate the objective to obtain the minimization form:

$$\min -\sum_{i \in V} x_i. \tag{2}$$

Step 2 — The feasibility condition requires that for any pair of vertices $\{i, j\}$ that are not connected by an edge (i.e., $\{i, j\} \notin E$), they cannot both be selected. This yields the inequality constraint:

$$x_i + x_j \leq 1, \quad \forall \{i, j\} \notin E. \tag{3}$$

Step 3 — To incorporate this inequality into the objective, we construct a penalty term. Because $x_i, x_j \in \{0, 1\}$, the product $x_i x_j$ is 1 if and only if both variables are 1, and 0 otherwise. Thus, the term $\lambda x_i x_j$ correctly penalizes only the infeasible configuration $(1, 1)$ while leaving $(0, 0)$, $(1, 0)$, and $(0, 1)$ unpenalized. We explicitly avoid the equality penalty $(x_i + x_j - 1)^2$, as it evaluates to 1 for both $(0, 0)$ and $(1, 1)$, incorrectly penalizing valid configurations where neither vertex is selected. The total penalty contribution is:

$$\lambda \sum_{\{i,j\} \notin E} x_i x_j. \tag{4}$$

Step 4 — Combining the objective and penalty terms gives the single QUBO Hamiltonian:

$$H(\mathbf{x}) = -\sum_{i\in V} x_i + \lambda \sum_{\{i,j\}\notin E} x_i x_j. \tag{5}$$

Step 5 — Reading off the coefficients for the standard QUBO form $H(\mathbf{x}) = \sum_i Q_{i,i}x_i + \sum_{i<j} Q_{i,j}x_i x_j + c$, we obtain the following closed-form expressions for the matrix entries and offset:

$$Q_{i,i} = -1, \quad \forall i \in V, \tag{6}$$
$$Q_{i,j} = \lambda, \quad \forall \{i,j\} \notin E, \tag{7}$$
$$Q_{i,j} = 0, \quad \forall \{i,j\} \in E, \tag{8}$$
$$c = 0. \tag{9}$$

Penalty Weight Justification. The penalty weight λ must be chosen large enough to ensure that any violation of the clique constraints results in a strictly higher energy than the best possible feasible solution. Violating a single constraint $x_i + x_j \leq 1$ corresponds to adding at most one extra vertex to the clique, which increases the objective sum by at most 1. Therefore, any $\lambda > 1$ guarantees that the energy cost of the penalty outweighs the maximum possible gain from the objective. We set $\lambda = 2$ to satisfy this condition with a safe margin.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^N$ recovers the maximum clique. For any feasible solution (a valid clique), all non-adjacent pairs have at least one variable equal to 0, making every penalty term $x_i x_j$ equal to 0. Thus, $H(\mathbf{x}) = -|C|$, and minimizing H is equivalent to maximizing $|C|$. For any infeasible solution, at least one pair $\{i,j\} \notin E$ has $x_i = x_j = 1$, contributing at least λ to the energy. Since $\lambda > 1$ and the maximum possible objective gain from adding a vertex is 1, the energy of any infeasible configuration strictly exceeds that of the optimal feasible configuration. Consequently, the global minimum of H must be feasible and correspond to a maximum clique.

Illustrative Example. Consider a graph with $N = 3$ vertices $V = \{0,1,2\}$ and no edges, i.e., $E = \emptyset$. Here, every pair is a non-edge. Setting $\lambda = 2$, the general formulas instantiate as follows. The objective terms are $-x_0 - x_1 - x_2$. The penalty terms are $2x_0x_1 + 2x_0x_2 + 2x_1x_2$. The resulting Hamiltonian is:

$$H(\mathbf{x}) = -x_0 - x_1 - x_2 + 2x_0x_1 + 2x_0x_2 + 2x_1x_2. \tag{10}$$

Evaluating H over all $2^3 = 8$ binary assignments yields the minimum energy -1, achieved at configurations such as $(x_0, x_1, x_2) = (1,0,0)$, $(0,1,0)$, or $(0,0,1)$. These correspond to selecting exactly one vertex, which is the maximum clique size for an empty graph.